MODULE 31 · EXAM STRATEGY - ALL DOMAINS

How KCNA questions are written

The exam's format, the question styles, the distractor patterns, and how to use ninety minutes.

SECTION Exam skills COVERS All domains MODULE 31 of 35

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KCNA tests **concepts and choices**, not typing. Each question has one best answer among **plausible neighbours**.

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The exam at a glance

ITEM	DETAIL
Format	Online, remotely proctored, multiple choice, in a secure browser
Time	90 minutes
Questions	About 60, as widely reported. Read each question's instructions
Pass mark	75%
Results	Within 24 hours, scored automatically
Attempts and validity	Two attempts; the certification is valid for two years

ON THE EXAM

Check the Linux Foundation's exam page and candidate handbook the week you book: details can change.

Pacing ninety minutes

- **About 1.5 minutes per question** on average. Most need far less.
- **First pass:** answer what you know, flag anything that needs thought, move on.
- **Aim to finish the first pass** with about thirty minutes to spare.
- **Second pass:** the flagged questions, with the time you saved.
- **Never leave a question blank.** An unanswered question scores nothing.

> ON THE EXAM
>
> One stubborn question should never cost you three easy ones.

Question styles you will meet

STYLE	LOOKS LIKE	TESTS
Definition	What is a DaemonSet?	Vocabulary
Component	Which component stores cluster state?	Architecture
Command	Which command shows each Pod's node?	kubectl fluency
Scenario	A team needs X, without Y. What should they do?	Choosing between neighbours
Negative	Which is NOT a graduated project?	Careful reading
Ecosystem	Which project provides event-driven autoscaling?	CNCF landscape

ON THE EXAM

Newer questions lean towards scenarios and commands; fewer are pure definitions.

Reading the question

- **Read the last sentence first:** it tells you what is being asked.
- **Find the planted detail:** 'every node', 'without restarting', 'one namespace only'.
- **Mark the qualifiers:** BEST, MOST, FIRST, LEAST effort, NOT, EXCEPT.
- **List the constraints:** must, must not, cannot, only.

> ON THE EXAM
>
> A missed NOT turns a known answer into a wrong one. Read the stem twice.

True versus best

TRUE

Works, technically

- Would solve part of the problem
- May break a stated constraint
- May be more effort than needed
- Often the tempting distractor

BEST

Meets every constraint

- Solves the whole problem
- Respects every must and must not
- Least effort, least privilege
- The answer the exam wants

THE TIEBREAKER

Several options may be true. Only one satisfies every constraint in the question.

Six distractor patterns

PATTERN	EXAMPLE FROM THIS COURSE
The neighbour	Readiness offered when the answer is liveness
Right tool, wrong layer	kube-proxy offered for a DNS failure
The over-reach	cluster-admin, or Restricted when Baseline is asked for
The outdated fact	dockershim, PodSecurityPolicy, SMI, OpenTracing
The half answer	Cordon when the Pods must also move
The invented feature	RBAC deny rules; a day-long ConfigMap cache

ON THE EXAM

Name the pattern, and the distractor usually eliminates itself.

Eliminating to the answer

01

Remove invented

Features that do not exist.

02

Remove wrong layer

Right idea, wrong component.

03

Remove over-reach

More power or effort than asked.

04

Compare the last two

Reread the planted detail.

05

Commit

Change later only for a specific reason.

ON THE EXAM

Most questions come down to two options. The planted detail decides between them.

Command questions

- **Read commands as grammar:** verb, type, name, flags. Module 10.
- **Know the flags that matter:** -o wide, -n and -A, --previous, -c, --dry-run=client -o yaml, apply -k.
- **Watch for invented flags** and verbs that do the wrong thing, such as create instead of apply.
- **Short names are fair game:** po, svc, deploy, ns, cm, pvc, sts, ds.

> ON THE EXAM
>
> A command with one wrong flag is a wrong answer. Read every flag.

Absolute words: handle with care

- **'Always', 'never', 'only'** are often wrong in a flexible ecosystem.
- **But not always:** 'only the API server talks to etcd' is true.
- **And 'never newer than the API server'** is exactly the kubelet skew rule.
- **Test the absolute against a fact you know,** rather than rejecting it on sight.

> ON THE EXAM
>
> Absolutes are a clue, not a verdict. Check them against the facts.

Exam day

- **Run the system check** for the secure browser well before the day.
- **A clear desk, a single monitor, valid ID,** and a quiet, private room.
- **Start early** enough to allow the identity and room checks.
- **If something breaks,** contact the proctor through the exam's chat; do not just leave.

> ON THE EXAM
>
> Most exam-day failures are technical, not knowledge. Remove them in advance.

The last week

- **Take the four practice exams** in this course under timed conditions.
- **Review every wrong answer by domain,** and rewatch that module's scenarios.
- **Replay modules 29 and 30:** what changed, and the confusion pairs.
- **Check cncf.io/projects** for any maturity changes since this course.

> ON THE EXAM
>
> Aim for a consistent 85% or more in practice before booking. That leaves margin above 75%.

> This question tests careful reading of a negative stem.

Which of these is NOT a CNCF graduated project?

A Prometheus

B Envoy

C Grafana

D Helm

◆ PAUSE AND DECIDE

Which of these is NOT a CNCF graduated project?

A Prometheus

Graduated: the CNCF's second.

B Envoy

Graduated.

C Grafana

A Grafana Labs project, not CNCF at all.

D Helm

Graduated.

THE SIGNAL

NOT questions: find the one that fails the test, not the ones that pass.

> A new booking API Pod has been Pending for ten minutes. The on-call engineer does not yet know why.

What should the engineer do FIRST?

A Run kubectl describe pod on it

B Add more nodes to the cluster

C Delete the Pod so it is recreated

D Restart the kubelet on every node

◆ PAUSE AND DECIDE

What should the engineer do FIRST?

A Run kubectl describe pod on it

Read the cause before changing anything.

B Add more nodes to the cluster

Might help, if the cause is capacity. Unknown yet.

C Delete the Pod so it is recreated

A replacement hits the same cause.

D Restart the kubelet on every node

Drastic, and unrelated unless nodes are the issue.

THE SIGNAL

FIRST means diagnose before you act.

> Developers need to view Pods, Services and Deployments in the dev namespace only. Nothing else, and no changes.

Which grant follows least privilege?

A cluster-admin through a ClusterRoleBinding

B The edit ClusterRole through a RoleBinding in dev

C The view ClusterRole through a ClusterRoleBinding

D The view ClusterRole through a RoleBinding in dev

◆ PAUSE AND DECIDE

Which grant follows least privilege?

A cluster-admin through a ClusterRoleBinding

The ultimate over-reach.

B The edit ClusterRole through a RoleBinding in dev

Right scope, but allows changes.

C The view ClusterRole through a ClusterRoleBinding

Read-only, but in every namespace.

D The view ClusterRole through a RoleBinding in dev

Read-only, in dev only.

THE SIGNAL

The over-reach pattern: pick the least power that meets the need.

> A ServiceAccount has a RoleBinding to the edit ClusterRole in the quayside namespace. It must stop being able to delete Pods.
>
> It should keep its other permissions where possible.

What should the administrator do?

A Add an RBAC rule that denies delete on Pods

B Replace the binding with a narrower custom Role

C Apply a NetworkPolicy to the ServiceAccount

D Create a PodDisruptionBudget for the Pods

◆ PAUSE AND DECIDE

What should the administrator do?

A Add an RBAC rule that denies delete on Pods

Invented: RBAC has no deny rules.

B Replace the binding with a narrower custom Role

RBAC only adds; remove the power by granting less.

C Apply a NetworkPolicy to the ServiceAccount

Network rules do not govern API permissions.

D Create a PodDisruptionBudget for the Pods

Limits voluntary evictions, not API deletes.

THE SIGNAL

The invented-feature pattern: RBAC deny rules do not exist.

Module 31 in five lines

- **90 minutes, 75% to pass,** about 60 questions; read each question's instructions.
- **Two passes:** answer and flag, then return with the time saved.
- **Last sentence first,** then the planted detail and the qualifiers.
- **True is not best.** The best option meets every constraint.
- **Name the distractor:** neighbour, wrong layer, over-reach, outdated, half, invented.

Now test yourself: four practice exams, twenty questions each, weighted like the real exam.